MOHAMMAD KHATERI, PH.D. CANDIDATE

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Permanent Residence: Finland

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EDUCATION

University of Eastern Finland, Finland *2020–2026*
Ph.D. in Computational Imaging & AI *(thesis submission: June 2026)*
Thesis: "Computational Methods for Brain Ultrastructure Analysis: Simulation, Super-Resolution, and Segmentation"; conducting multidisciplinary research integrating computer science, computational imaging, and medical physics. Under the supervision of Prof. Alejandra Sierra and Prof. Jussi Tohka.

Harvard Medical School, USA *Dec 2024–Jun 2025*
Visiting Scholar
Conducted research on AI-driven imaging methods at Harvard Medical School in groups led by Prof. Simon K. Warfield and Prof. P. Ellen Grant, focusing on MRI super-resolution and ultrasound image compounding.

Tarbiat Modares University, Iran *2014–2017*
Master of Science in Electrical Engineering–Communications
Thesis: Multi-sensor image fusion based on spatial-frequency features.
Under the supervision of Prof. Hassan Ghassemian.

Bu-Ali Sina University, Iran *2009–2013*
Bachelor of Science in Electrical Engineering–Electronics
Thesis: Examining electrical properties of concrete to analyze its physical properties.

RESEARCH PROJECTS

Weakly Supervised Learning for Ultrastructural Segmentation in Electron Microscopy *2024–present*
University of Eastern Finland, Finland
Developing weakly supervised learning frameworks for soma instance segmentation in large-scale 3D electron microscopy images under severe annotation scarcity.

MRI Super-Resolution *2025–present*
Harvard Medical School, USA
Developing deep learning methods for reconstructing high-resolution isotropic MRI from fast multi-view anisotropic acquisitions, addressing clinical constraints on scan time and spanning supervised, self-supervised, and scan-specific learning paradigms.

Computational Imaging for Ultrasound *2025–2026*
Harvard Medical School, USA
Developed multi-view ultrasound compounding methods for fetal brain imaging to mitigate view-dependent artifacts and limited field of view, implementing and evaluating classical and learning-based strategies with a focus on unsupervised and self-supervised (e.g., Plug-and-Play) approaches, released as the USFetal Compounding Toolbox.

Image Restoration for Large-Scale Electron Microscopy *2022–2024*
University of Eastern Finland, Finland
Developed supervised and self-supervised pipelines for in-place super-resolution and isotropic 3D reconstruction of large-scale electron microscopy images from anisotropic acquisitions, along with efficient denoising methods to enhance image quality.

Simulation of Diffusion MRI with Parallel Computing *2020–2022*
University of Eastern Finland, Finland
Developed biophysically grounded Monte Carlo simulations of diffusion MRI using CUDA-based parallel computing on HPC infrastructure, including simulations of advanced pulse sequences.

Multi-Sensor Image Fusion *2016–2019*
Tarbiat Modares University, Iran; University of British Columbia, Canada
Developed model-based and learning-based pipelines for fusing remote-sensing data (panchromatic, multispectral, and hyperspectral) and MR–PET images, leveraging multi-resolution analysis, compressed sensing, and sparse representations.

Image Quality Assessment *2016–2018*
Tarbiat Modares University, Iran
Evaluated image sharpness and interpretability using Modulation Transfer Function analysis and task-driven quality metrics in remote-sensing applications.

WORK EXPERIENCE

Tarbiat Modares University & University of British Columbia *2017–2019*
Researcher – Conducted research on image processing and multi-sensor data fusion.

Huawei Technologies *2016–2017*
RF Engineer – Conducted wireless communications quality assessment.

RESEARCH INTERESTS

Image Processing & Analysis	Low-Level Vision
Computational Imaging & Inverse Problems	Generative AI & Image Synthesis
Vision Foundation Models	Image & Video Enhancement

PROGRAMMING AND TECHNICAL SKILLS

Programming: Python, C/C++, CUDA
Deep Learning Frameworks: PyTorch, MONAI, scikit-learn
Computer Vision & AI: CNNs, Vision Transformers (ViT, Swin), GANs, Diffusion Models, Foundation Models
Scientific Computing: NumPy, SciPy, OpenCV, Matplotlib, Jupyter
Infrastructure: Linux, Docker, Git/GitHub, HPC, GPU Computing (NVIDIA, AMD)
Mathematics: Optimization, Numerical Analysis, Probability & Statistics, Signal Processing

AWARDS AND HONORS

CVPR Highlight (co-author of H-ViT)	2024
Best Paper Runner-Up, *EUVIP Workshop*	2023
Featured Cover Image in *NMR in Biomedicine*	2022
Oral Presentation at ISMRM (acceptance rate 12%)	2022
Top 1% Rank in National Entrance Exam for M.Sc.	2014
Outstanding Student Award (1st Rank) in B.Sc. in Electrical Engineering	2013

GRANTS

Doctoral Researcher Position (competitively awarded by UEF, salary-funded)	2022–2025
Finnish Cultural Foundation Grant (€15,000)	2025–2026
Finnish Foundation for Technology Promotion Grant (€13,200)	2025–2026
Saastamoinen Foundation Travel Grant (€12,000)[†]	2024–2025
KAUTE Foundation Grant (€5,700)[†]	2024–2025

DPMM Travel Grants (€1,000)†	2023
DPMM Travel Grants (€600)	2022

†*Awarded to support a six-month research visit to Harvard Medical School and Boston Children's Hospital, 2024–2025.*

SELECTED COURSES, WORKSHOPS, AND CONFERENCES

Deep Neural Networks and Computer Vision	2022
CUDA C/C++ Programming	2021
Pattern Recognition	2021
Workshop on Advanced MRI Methods, Finland	2022
International Society for Magnetic Resonance in Medicine (ISMRM), London, UK	2022
Medical Image Computing Summer School, University College London, UK	2022
European Workshop on Visual Information Processing (EUVIP), Gjøvik, Norway	2023
CITI Program: Human Subjects Research & HIPAA Training, Harvard Medical School, USA	2025

REVIEWER FOR ACADEMIC JOURNALS AND CONFERENCES

Journals: IEEE Trans. Neural Networks and Learning Systems; IEEE Trans. Image Processing; IEEE Trans. Medical Imaging; IEEE J. Biomedical and Health Informatics; IEEE J. Selected Topics in Signal Processing; IEEE Signal Processing Letters; Pattern Recognition Letters; and 9+ others.

Conferences: CVPR; ECCV; NeurIPS; ICPR.

LANGUAGES

English: Proficient
Persian: Native

MEMBERSHIPS

Student Member, Institute of Electrical and Electronics Engineers (IEEE)
Trainee Member, International Society for Magnetic Resonance in Medicine (ISMRM)

PUBLICATIONS

Author of fourteen journal and conference papers, as listed below; the complete list is available on my **Google Scholar**.

Journal Articles

- **M. Khateri**, M. Ghahremani, Sergio Valencia, Camilo Jaimes, A. Sierra, J. Tohka, P. E. Grant, and D. Karimi, "USFetal: Tools for Fetal Brain Ultrasound Compounding," *IEEE Journal of Biomedical and Health Informatics* (submitted), 2026. [GitHub]
- **M. Khateri**, S. Vasylechko, M. Ghahremani, L. Timms, D. Kocanaogullari, S. K. Warfield, C. Jaimes, D. Karimi, A. Sierra, J. Tohka, S. Kurugol, and O. Afacan, "MRI Super-Resolution with Deep Learning: A Comprehensive Survey," *Medical Image Analysis* (submitted), 2026. [GitHub]
- **M. Khateri**, M. Ghahremani, A. Sierra, and J. Tohka, "No-Clean-Reference Image Super-Resolution: Application to Electron Microscopy," *IEEE Transactions on Computational Imaging*, 2024. [GitHub]
- M. Ghahremani, **M. Khateri**, A. Sierra, and J. Tohka, "Adversarial Distortion Learning for Medical Image Denoising," *arXiv preprint*. [GitHub]
- **M. Khateri**, M. Reisert, A. Sierra, J. Tohka, and V. G. Kiselev, "What Does FEXI Measure?" *NMR in Biomedicine*, vol. 35, no. 12, e4804, 2022. (Featured on the cover of vol. 35, no. 12)

- **M. Khateri**, F. Shabanzade, F. Mirzapour, A. Zaji, and Z. Liu, "A Variational Approach for Fusion of Panchromatic and Multispectral Images Using a New Spatial–Spectral Consistency Term," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 3421–3436, 2020.
- **M. Khateri**, F. Shabanzade, and F. Mirzapour, "Regularised IHS-Based Pan-Sharpening Approach Using Spectral Consistency Constraint and Total Variation," *IET Image Processing*, vol. 14, no. 1, pp. 94–104, 2019.
- F. Shabanzade, **M. Khateri**, and Z. Liu, "MR and PET Image Fusion Using Nonparametric Bayesian Joint Dictionary Learning," *IEEE Sensors Letters*, vol. 3, no. 7, pp. 1–4, 2019.

Under Preparation

- **M. Khateri**, M. Ghahremani, A. Sierra, and J. Tohka, "SomaSeg: Weakly-Supervised Learning for Soma Segmentation in 3D Electron Microscopy," *IEEE Transactions on Biomedical Engineering* (to be submitted May 2026).
- **M. Khateri**, D. Kocanaogullari, M. Ghahremani, A. Sierra, J. Tohka, S. K. Warfield, and O. Afacan, "MRI-SR: Supervised and Self-Supervised Learning for Multi-View MRI Super-Resolution," *IEEE Signal Processing Letters* (to be submitted July 2026).

Conference Proceedings

- B. Li, S. Valencia, C. Jaimes, **M. Khateri**, A. Taymourtash, E. Grant, S. Warfield, and D. Karimi, "Fiber Orientation Mapping in the Developing Brain via Unsupervised Learning-Based Spherical Deconvolution," *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2026 (submitted).
- M. Ghahremani, **M. Khateri**, B. Jian, B. Wiestler, E. Adeli, and C. Wachinger, "H-ViT: A Hierarchical Vision Transformer for Deformable Image Registration," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. (Highlights)
- **M. Khateri**, M. Ghahremani, A. Sierra, and J. Tohka, "Self-Supervised Super-Resolution Approach for Isotropic Reconstruction of Three-Dimensional Electron Microscopy Images from Anisotropic Acquisition," *11th European Workshop on Visual Information Processing (EUVIP)*, 2023. (Oral, Best Paper Runner-Up)
- **M. Khateri**, M. Reisert, A. Sierra, J. Tohka, and V. G. Kiselev, "What Does FEXI Measure?" *International Society for Magnetic Resonance in Medicine (ISMRM)*, London, UK, 2022. (Oral)
- **M. Khateri**, H. Ghassemian, and F. Mirzapour, "A Model-Based Method for Pan-Sharpening of Multi-Spectral Images Using Sparse Representation," *IEEE International Conference on Signal and Image Processing Applications (ICSIPA)*, pp. 219–224, 2019. (Oral)
- **M. Khateri** and H. Ghassemian, "A Self-Learning Approach for Pan-Sharpening of Multispectral Images," *IEEE International Conference on Signal and Image Processing Applications (ICSIPA)*, pp. 199–204, 2017. (Oral)